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```
% Author: Jiale Jiang
% Editor: Jiale Jiang(self-edited)
```

1. Data Parsing

```
% a.
% load data from previously saved workspace
% the data was imported as numerical matrix, following data parsing can
% help our code to be way easier to work with and more human readable;

load("CarOBD2data.mat", "Jetta1", "Jetta2", "Jetta3","Jetta5PID1",...
    "Jetta5PID2","Jetta5PID3");

% b.
% create a cell containing 3 raw data matricies
RawJettaData = {Jetta1, Jetta2, Jetta3};

% create a vector of string names for struct name parsing
newNameJetta = ["Time_sec", "TripDistance_mile", "VehicleSpeed_MPH", ...
    "Acceleration_fts2", "TripFuel_gal", "TotalFuelEcon_MPG", ...
    "InstantFuelEcon_MPG", "EngineRPM_rpm"];
% c.
% parse raw data into two cells;
JettaData = parsingData(newNameJetta,RawJettaData);
```

2. Compute Data Sampling Frequency

```
% a. Frequency calculation
JettaTimeData = cell(1,3);

for i = 1:3
    JettaTimeData{i} = calculateTimeData(JettaData{i}.Time_sec);
end
```

3. Data Presentation as table output in MATLAB terminal

```
% a. 2022 Volkswagen Jetta SE;
% Test Procedure: 1)Park at a relatively flat parking lot;
```

```
%                   2)Accelerate the car from 0-35 mph;
%                   3)Once reach 35 mph,deccelerate the car to 0 mph safely;

printJettaDataTable(JettaData);
```

```
Jetta Trial 1
Time(s)  Speed(mph)  Accel(ft/s^2)  RPM   TripDist(mi)  InstMPG  TripFuel  TotalMPG
0.091      0.00          0.000      875.0    0.14918     0.00      0.00      0.00
0.244      0.00          0.000      875.0    0.14918     0.00      0.01      9.49
0.598      0.00          0.000      872.0    0.14918     0.00      0.01      9.49
0.669      0.00          0.000      875.5    0.14918     0.00      0.01      9.49
0.738      0.00          0.000      875.5    0.14918     0.00      0.01      9.49
0.994      0.00          0.000      875.5    0.14918     0.00      0.01      9.49
1.201      0.00          0.000      882.5    0.14918     0.00      0.01      9.49
1.392      0.00          0.000      877.0    0.14918     0.00      0.01      9.49
1.451      0.00          0.000      896.5    0.14918     0.00      0.01      9.49
1.509      0.00          0.000      896.5    0.14918     0.00      0.01      9.48
1.579      0.00          0.000      896.5    0.14918     0.00      0.01      9.48
1.687      0.00          0.000     1313.0    0.14918     0.00      0.01      9.48
1.867      1.86         19.313     1731.5    0.14922     1.78      0.01      9.48
1.937      2.49         25.762     1731.5    0.14927     1.75      0.01      9.48
2.193      3.11          8.231     1731.5    0.14949     2.07      0.01      9.47
2.297      4.97         14.913     2027.5    0.14963     2.88      0.01      9.47
2.397      6.21         22.287     2027.5    0.14980     2.88      0.01      9.47
2.431      6.84         15.824     2145.0    0.14993     2.90      0.01      9.46
2.521      7.46         14.823     2145.0    0.15005     2.90      0.01      9.46
2.556      8.08         14.602     2145.0    0.15021     2.94      0.01      9.46
2.664      8.70         12.742     2380.5    0.15038     2.99      0.01      9.45
2.721      9.32         14.317     2380.5    0.15053     2.67      0.01      9.45
2.789      9.32          8.022     2380.5    0.15071     2.50      0.01      9.45
2.871      9.94          5.505     2560.5    0.15094     2.49      0.01      9.44
2.930     10.56         13.350     2560.5    0.15111     2.37      0.01      9.44
3.000     11.18         14.299     2560.5    0.15133     2.48      0.01      9.43
3.071     11.18          6.454     2949.5    0.15154     2.46      0.01      9.42
3.139     11.81          6.631     2949.5    0.15177     2.23      0.02      9.42
3.395     13.05         10.199     2949.5    0.15270     2.48      0.02      9.40
3.498     14.29         12.425     2587.0    0.15310     2.48      0.02      9.40
3.549     14.91         13.263     2587.0    0.15353     3.41      0.02      9.38
3.659     15.53         12.227     2705.0    0.15378     3.37      0.02      9.38
3.726     16.16         14.627     2705.0    0.15408     3.44      0.02      9.37
3.992     16.78          8.522     2705.0    0.15532     3.65      0.02      9.36
4.099     18.64         14.533     2920.5    0.15587     3.65      0.02      9.36
4.146     19.88         22.456     2920.5    0.15640     4.37      0.02      9.35
4.255     20.51         17.013     2968.0    0.15675     4.59      0.02      9.34
4.815     20.51          7.374     2968.0    0.15994     4.57      0.02      9.31
4.991     22.99         10.367     2776.5    0.16106     4.92      0.02      9.31
5.060     24.23         23.588     2707.0    0.16153     5.62      0.02      9.30
5.301     24.23         13.221     2707.0    0.16315     5.80      0.02      9.30
5.501     25.48          4.537     2803.0    0.16457     6.16      0.02      9.29
5.593     26.72         14.470     2788.0    0.16525     6.14      0.02      9.29
5.651     27.34         17.812     2848.5    0.16569     6.29      0.02      9.28
5.709     27.96         15.782     2848.5    0.16614     6.29      0.02      9.28
5.792     28.58         10.628     2871.5    0.16746     6.37      0.02      9.28
```

6.092	28.58	2.725	2871.5	0.16918	6.32	0.02	9.27
6.197	29.83	8.674	2883.0	0.17005	6.56	0.02	9.27
6.264	29.83	8.674	2928.0	0.17060	6.50	0.02	9.27
6.320	29.83	0.000	2928.0	0.17107	6.37	0.02	9.26
6.389	30.45	6.612	2928.0	0.17165	6.41	0.02	9.26
6.593	30.45	6.612	2723.0	0.17338	6.41	0.02	9.26
6.698	31.69	8.688	2600.5	0.17430	7.32	0.02	9.25
6.793	32.31	13.473	2600.5	0.17515	7.81	0.02	9.25
6.850	32.31	4.785	2524.5	0.17566	7.81	0.02	9.25
6.918	32.31	0.000	2524.5	0.17628	8.01	0.02	9.25
6.987	32.93	6.617	2524.5	0.17691	8.16	0.02	9.25
7.193	32.93	6.617	2505.5	0.17879	8.16	0.02	9.25
7.251	33.55	7.791	2498.0	0.17933	8.46	0.02	9.25
7.309	34.18	15.654	2498.0	0.17988	8.73	0.02	9.25
7.491	34.18	7.863	2498.0	0.18161	8.86	0.02	9.25
7.550	34.80	7.784	2404.0	0.18218	13.13	0.02	9.25
7.619	34.80	7.784	2404.0	0.18285	16.57	0.02	9.25
7.797	34.80	0.000	2230.5	0.18457	20.40	0.02	9.26
7.867	34.18	-6.541	2195.0	0.18523	33.87	0.02	9.26
8.101	34.18	-6.541	2195.0	0.18746	42.21	0.02	9.28
8.194	33.55	-4.910	2138.0	0.18832	69.57	0.02	9.28
8.298	32.93	-9.279	2060.5	0.18927	75.49	0.02	9.29
8.403	32.93	-4.368	2060.5	0.19023	84.92	0.02	9.30
8.461	32.93	0.000	2077.0	0.19076	92.30	0.02	9.30
8.519	32.93	0.000	2077.0	0.18641	113.19	0.02	9.26
8.589	32.31	-6.555	2077.0	0.18704	129.66	0.02	9.27
8.660	32.31	-6.555	2032.5	0.18767	151.52	0.02	9.27
8.731	31.07	-12.875	2032.5	0.18828	148.88	0.02	9.28
9.143	31.69	-11.769	2032.5	0.19191	157.93	0.02	9.31
9.201	30.45	-14.419	1964.5	0.19241	151.74	0.02	9.31
9.260	29.83	-23.369	1924.0	0.19289	153.79	0.02	9.31
9.327	29.83	-7.844	1924.0	0.19345	157.02	0.02	9.32
9.419	29.83	0.000	1924.0	0.19421	157.02	0.02	9.32
9.479	29.20	-7.585	1892.0	0.19470	153.75	0.02	9.33
9.546	29.20	-7.585	1892.0	0.19524	156.35	0.02	9.33
9.615	29.20	0.000	1892.0	0.19580	156.85	0.02	9.34
9.670	29.20	0.000	1842.0	0.19625	156.85	0.02	9.34
9.739	28.58	-6.661	1842.0	0.19679	157.68	0.02	9.34
9.809	28.58	-6.661	1842.0	0.19735	157.68	0.02	9.35
9.869	27.96	-7.588	1799.5	0.19782	154.25	0.02	9.35
9.940	27.96	-7.588	1799.5	0.19837	157.89	0.02	9.36
10.093	27.96	0.000	1799.5	0.19956	157.89	0.02	9.37
10.151	27.34	-7.851	1753.0	0.20000	154.39	0.02	9.37
10.209	26.72	-15.768	1753.0	0.20043	154.88	0.02	9.37
10.503	26.10	-9.470	1729.0	0.20256	151.28	0.02	9.39
10.560	25.48	-9.507	1670.5	0.20296	149.72	0.02	9.39
10.620	26.10	-0.364	1670.5	0.20340	158.75	0.02	9.40
10.690	25.48	1.139	1621.0	0.20390	154.97	0.02	9.40
10.757	24.85	-13.348	1621.0	0.20435	155.81	0.02	9.40
10.997	24.85	-6.897	1621.0	0.20601	155.81	0.02	9.42
11.094	23.61	-9.406	1626.0	0.20665	154.18	0.02	9.42
11.152	23.61	-9.406	1626.0	0.20703	154.19	0.02	9.42
11.393	22.99	-1.888	1626.0	0.20857	150.14	0.02	9.43
11.450	21.75	-18.002	1818.0	0.20891	136.34	0.02	9.44

11.697	21.13	-17.957	1818.0	0.21036	118.46	0.02	9.45
11.812	20.51	-5.790	1782.5	0.21102	110.55	0.02	9.45
11.880	19.26	-17.408	1736.0	0.21138	110.16	0.02	9.46
12.113	18.64	-15.421	1736.0	0.21259	109.46	0.02	9.47
12.170	19.26	6.054	1658.5	0.21289	113.11	0.02	9.47
12.239	18.64	1.480	1658.5	0.21325	114.57	0.02	9.47
12.311	18.02	-12.927	1572.0	0.21361	110.75	0.02	9.47
12.369	18.02	-6.392	1572.0	0.21390	116.85	0.02	9.48
12.440	18.02	0.000	1572.0	0.21426	116.85	0.02	9.48
12.502	18.02	0.000	1518.5	0.21426	116.85	0.02	9.48
12.532	17.40	-7.659	1518.5	0.21485	116.79	0.02	9.48
12.620	17.40	-7.659	1518.5	0.21514	116.79	0.02	9.48
12.800	16.78	-2.521	1453.0	0.21598	112.62	0.02	9.49
12.883	16.16	-8.007	1351.5	0.21635	113.34	0.02	9.49
12.940	16.16	-5.486	1351.5	0.21661	121.85	0.02	9.50
13.192	15.53	-1.807	1351.5	0.21769	117.17	0.02	9.50
13.250	14.29	-17.572	1122.5	0.21792	103.65	0.02	9.51
13.321	14.29	-15.766	1122.5	0.21820	125.18	0.02	9.51
13.500	14.29	0.000	1122.5	0.21891	125.18	0.02	9.51
13.694	13.05	-4.675	1084.5	0.21962	123.82	0.02	9.52
13.912	11.81	-8.857	996.5	0.22033	121.00	0.02	9.52
13.974	11.18	-11.611	910.5	0.22052	110.36	0.02	9.53
14.030	11.18	-7.429	910.5	0.22070	83.07	0.02	9.53
14.297	10.56	-1.709	870.5	0.22148	67.94	0.02	9.53
14.520	9.94	-3.754	867.0	0.22210	59.36	0.02	9.54
14.703	8.70	-7.018	846.0	0.22254	36.88	0.02	9.54
14.895	8.08	-7.347	855.0	0.22297	27.41	0.02	9.54
14.953	7.46	-10.202	838.5	0.22309	20.41	0.02	9.54
15.019	6.84	-14.729	838.5	0.22322	17.84	0.02	9.54
15.190	6.84	-6.901	839.0	0.22354	17.74	0.02	9.54
15.261	6.21	-6.457	835.0	0.22366	16.04	0.02	9.54
15.492	6.21	-6.457	835.0	0.22406	15.67	0.02	9.54
15.550	4.97	-15.741	854.5	0.22414	12.51	0.02	9.55
15.617	4.97	-15.741	854.5	0.22424	12.47	0.02	9.55
15.719	4.97	0.000	851.5	0.22438	12.46	0.02	9.55
15.784	4.35	-6.950	851.5	0.22446	10.94	0.02	9.55
15.842	4.35	-6.950	844.5	0.22453	10.71	0.02	9.55
15.909	3.73	-6.829	844.5	0.22459	9.07	0.02	9.55
15.977	3.73	-6.829	844.5	0.22467	9.07	0.02	9.55
16.198	3.11	-2.067	852.5	0.22486	7.55	0.02	9.55
16.320	2.49	-5.781	848.0	0.22494	6.11	0.02	9.54
16.390	2.49	-3.713	848.0	0.22499	6.14	0.02	9.54
16.456	1.86	-6.921	854.0	0.22502	4.51	0.02	9.54
16.524	1.86	-6.921	854.0	0.22506	4.48	0.02	9.54
16.701	1.86	0.000	854.0	0.22515	4.48	0.02	9.54
16.770	1.24	-6.533	866.0	0.22517	3.10	0.02	9.54
16.830	1.24	-6.533	866.0	0.22519	3.06	0.02	9.54
17.096	0.62	-1.715	859.0	0.22524	3.06	0.02	9.54
17.422	0.62	-1.715	873.5	0.22535	1.58	0.02	9.54
18.000	0.00	-1.818	884.5	0.22535	0.00	0.02	9.53
18.199	0.00	-1.818	878.5	0.22535	0.00	0.02	9.53
18.417	0.00	0.000	873.5	0.22535	0.00	0.02	9.53
18.596	0.00	0.000	876.0	0.22535	0.00	0.02	9.53
18.799	0.00	0.000	880.5	0.22535	0.00	0.02	9.53

18.871	0.00	0.000	882.0	0.22535	0.00	0.02	9.53
18.929	0.00	0.000	882.0	0.22535	0.00	0.02	9.53

Jetta Trial 2

Time(s)	Speed(mph)	Accel(ft/s^2)	RPM	TripDist(mi)	InstMPG	TripFuel	TotalMPG
0.077	0.00	0.000	870.0	0.34926	0.00	0.00	0.00
0.105	0.00	0.000	876.5	0.34926	0.00	0.04	8.93
0.204	0.00	0.000	876.5	0.34926	0.00	0.04	8.93
0.271	0.00	0.000	876.5	0.34926	0.00	0.04	8.93
0.341	0.00	0.000	876.0	0.34926	0.00	0.04	8.93
0.575	0.00	0.000	876.0	0.34926	0.00	0.04	8.93
0.633	0.00	0.000	871.5	0.34926	0.00	0.04	8.92
0.875	0.00	0.000	871.5	0.34926	0.00	0.04	8.92
0.934	0.00	0.000	879.0	0.34926	0.00	0.04	8.92
1.182	0.00	0.000	879.0	0.34926	0.00	0.04	8.92
1.287	0.00	0.000	882.0	0.34926	0.00	0.04	8.92
1.378	0.00	0.000	874.5	0.34926	0.00	0.04	8.92
1.436	0.00	0.000	874.5	0.34926	0.00	0.04	8.92
1.504	0.00	0.000	874.5	0.34926	0.00	0.04	8.92
1.563	0.00	0.000	873.5	0.34926	0.00	0.04	8.92
1.632	0.00	0.000	873.5	0.34926	0.00	0.04	8.92
1.724	0.00	0.000	873.5	0.34926	0.00	0.04	8.92
1.796	0.00	0.000	872.5	0.34926	0.00	0.04	8.92
1.985	0.00	0.000	872.5	0.34926	0.00	0.04	8.92
2.080	0.00	0.000	874.5	0.34926	0.00	0.04	8.92
2.183	0.00	0.000	879.5	0.34926	0.00	0.04	8.92
2.280	0.00	0.000	879.5	0.34926	0.00	0.04	8.92
2.374	0.00	0.000	877.0	0.34926	0.00	0.04	8.92
2.827	0.00	0.000	877.0	0.34926	0.00	0.04	8.91
2.886	0.00	0.000	876.5	0.34926	0.00	0.04	8.91
2.955	0.00	0.000	878.0	0.34926	0.00	0.04	8.91
3.025	0.00	0.000	878.0	0.34926	0.00	0.04	8.91
3.277	0.00	0.000	878.0	0.34926	0.00	0.04	8.91
3.336	0.00	0.000	870.5	0.34926	0.00	0.04	8.91
3.601	0.00	0.000	870.5	0.34926	0.00	0.04	8.91
3.780	0.00	0.000	881.0	0.34926	0.00	0.04	8.91
3.986	0.00	0.000	875.0	0.34926	0.00	0.04	8.91
4.180	0.00	0.000	873.0	0.34926	0.00	0.04	8.91
4.386	0.00	0.000	879.0	0.34926	0.00	0.04	8.90
4.481	0.00	0.000	872.0	0.34926	0.00	0.04	8.90
4.550	0.00	0.000	875.5	0.34926	0.00	0.04	8.90
4.782	0.00	0.000	875.5	0.34926	0.00	0.04	8.90
4.877	0.00	0.000	874.5	0.34926	0.00	0.04	8.90
4.935	0.00	0.000	879.0	0.34926	0.00	0.04	8.90
5.004	0.00	0.000	879.0	0.34926	0.00	0.04	8.90
5.073	0.00	0.000	879.0	0.34926	0.00	0.04	8.90
5.281	0.00	0.000	867.0	0.34926	0.00	0.04	8.90
5.387	0.00	0.000	872.0	0.34926	0.00	0.04	8.90
5.446	0.00	0.000	872.0	0.34926	0.00	0.04	8.90
5.677	0.00	0.000	872.0	0.34926	0.00	0.04	8.90
5.733	0.00	0.000	874.5	0.34926	0.00	0.04	8.90
5.986	0.00	0.000	874.5	0.34926	0.00	0.04	8.90
6.080	0.00	0.000	873.0	0.34926	0.00	0.04	8.89
6.188	0.00	0.000	874.0	0.34926	0.00	0.04	8.89

6.280	0.00	0.000	874.0	0.34926	0.00	0.04	8.89
6.347	0.00	0.000	865.5	0.34926	0.00	0.04	8.89
6.575	0.00	0.000	865.5	0.34926	0.00	0.04	8.89
6.633	0.00	0.000	875.5	0.34926	0.00	0.04	8.89
6.853	0.00	0.000	875.5	0.34926	0.00	0.04	8.89
7.075	0.00	0.000	874.5	0.34926	0.00	0.04	8.89
7.134	0.00	0.000	1029.0	0.34926	0.00	0.04	8.89
7.199	0.62	6.929	1029.0	0.34927	0.81	0.04	8.89
7.267	0.62	6.929	1029.0	0.34929	0.81	0.04	8.89
7.325	1.24	7.895	1534.0	0.34931	1.60	0.04	8.89
7.392	1.24	7.895	1534.0	0.34933	1.06	0.04	8.89
7.460	1.86	6.647	1534.0	0.34936	1.54	0.04	8.88
7.687	2.49	8.652	1867.5	0.34952	1.99	0.04	8.88
7.879	4.35	9.128	1990.5	0.34975	2.55	0.04	8.88
8.084	6.21	13.797	2088.5	0.35011	2.55	0.04	8.88
8.120	8.08	25.944	2239.0	0.35027	2.97	0.04	8.87
8.190	8.70	25.712	2239.0	0.35044	2.96	0.04	8.87
8.284	9.32	14.239	2239.0	0.35059	2.98	0.04	8.87
8.355	9.32	7.797	2485.5	0.35077	2.79	0.04	8.86
8.423	9.94	6.684	2485.5	0.35096	2.49	0.04	8.86
8.714	10.56	8.249	2485.5	0.35181	2.45	0.05	8.84
8.781	13.67	35.681	3254.0	0.35207	2.81	0.05	8.84
9.085	14.91	37.111	3254.0	0.35333	2.34	0.05	8.82
9.181	18.64	31.535	4352.0	0.35382	2.93	0.05	8.81
9.384	19.88	33.034	4352.0	0.35494	2.17	0.05	8.79
9.479	21.75	18.808	4842.5	0.35552	2.33	0.05	8.78
9.543	21.75	14.315	5218.0	0.35591	2.01	0.05	8.77
9.611	23.61	19.912	5218.0	0.35636	2.02	0.05	8.76
9.892	22.99	18.288	5218.0	0.35815	2.11	0.05	8.72
10.026	24.85	8.596	5584.5	0.35907	3.42	0.05	8.71
10.281	25.48	12.006	5584.5	0.36088	2.31	0.05	8.68
10.377	28.58	25.536	3924.5	0.36164	2.31	0.05	8.68
10.406	28.58	23.750	3924.5	0.37282	3.16	0.05	8.73
10.500	28.58	0.000	3924.5	0.37334	3.40	0.05	8.72
10.569	29.20	6.628	4025.5	0.37390	3.54	0.05	8.71
10.627	29.83	14.463	4025.5	0.37438	3.58	0.05	8.71
10.685	30.45	15.650	4025.5	0.37488	3.58	0.05	8.71
10.719	31.07	14.441	4254.0	0.37547	3.71	0.05	8.70
10.822	31.07	6.626	4254.0	0.37606	3.51	0.05	8.69
11.384	32.31	1.622	4254.0	0.38110	3.64	0.05	8.65
11.775	34.80	6.278	4119.0	0.38488	3.91	0.05	8.62
11.834	36.66	27.974	3179.0	0.38548	4.25	0.05	8.62
11.905	36.04	16.921	3179.0	0.38619	7.44	0.05	8.61
12.087	36.04	-6.397	3106.0	0.38802	9.16	0.05	8.62
12.485	34.80	-2.292	2901.0	0.39186	19.05	0.05	8.63
12.555	29.83	-54.051	2536.0	0.39244	45.76	0.05	8.63
12.613	29.83	-51.758	2536.0	0.39292	55.98	0.05	8.63
12.884	27.96	-5.046	2536.0	0.39503	55.98	0.05	8.63
12.929	25.48	-25.399	2115.0	0.39566	50.19	0.05	8.65
13.043	23.61	-39.946	2115.0	0.39612	109.77	0.05	8.65
13.285	22.99	-21.478	2115.0	0.39766	111.16	0.05	8.66
13.377	19.88	-26.559	1692.5	0.39817	96.14	0.05	8.66
13.434	18.02	-48.891	1692.5	0.39846	108.87	0.05	8.67
13.502	17.40	-30.887	1692.5	0.39879	105.45	0.05	8.67

13.572	17.40	-6.669	1480.0	0.39912	105.45	0.05	8.67
13.641	16.16	-13.276	1480.0	0.39943	111.97	0.05	8.67
13.882	14.91	-17.056	1480.0	0.40043	103.36	0.05	8.68
13.977	11.81	-27.688	1097.0	0.40074	75.77	0.05	8.68
14.036	11.18	-31.691	1097.0	0.40092	108.94	0.05	8.68
14.093	10.56	-15.653	1097.0	0.40109	102.89	0.05	8.68
14.164	9.94	-14.312	926.5	0.40129	92.97	0.05	8.68
14.381	9.32	-8.544	926.5	0.40185	83.22	0.05	8.69
14.487	7.46	-14.963	849.5	0.40207	42.12	0.05	8.69
14.581	5.59	-27.356	821.0	0.40222	20.15	0.05	8.69
14.651	5.59	-14.496	821.0	0.40232	17.14	0.05	8.69
14.878	4.35	-4.017	821.0	0.40260	13.09	0.05	8.69
14.936	3.11	-19.651	811.0	0.40265	9.21	0.05	8.69
15.007	2.49	-22.101	811.0	0.40270	6.64	0.05	8.69
15.065	2.49	-6.466	811.0	0.40274	6.31	0.05	8.69
15.135	1.86	-6.463	839.5	0.40277	4.77	0.05	8.69
15.475	1.86	-6.463	839.5	0.40295	4.95	0.05	8.69
15.533	0.62	-15.741	848.0	0.40296	4.95	0.05	8.69
15.568	0.62	-15.741	848.0	0.40297	1.58	0.05	8.69
15.671	0.00	-6.683	848.0	0.40297	0.00	0.05	8.69
15.735	0.00	-6.683	865.5	0.40297	0.00	0.05	8.69
15.802	0.00	0.000	865.5	0.40297	0.00	0.05	8.69
15.871	0.00	0.000	865.5	0.40297	0.00	0.05	8.69
15.934	0.00	0.000	872.0	0.40297	0.00	0.05	8.68
16.178	0.00	0.000	872.0	0.40297	0.00	0.05	8.68
16.236	0.00	0.000	896.5	0.40297	0.00	0.05	8.68
16.481	0.00	0.000	896.5	0.40297	0.00	0.05	8.68

Jetta Trial 3

Time(s)	Speed(mph)	Accel(ft/s^2)	RPM	TripDist(mi)	InstMPG	TripFuel	TotalMPG
0.090	0.00	0.000	877.0	0.47805	0.00	0.00	0.00
0.124	0.00	0.000	881.0	0.47805	0.00	0.07	8.53
0.385	0.00	0.000	881.0	0.47805	0.00	0.07	8.53
0.688	0.00	0.000	878.0	0.47805	0.00	0.07	8.53
0.759	0.00	0.000	880.5	0.47805	0.00	0.07	8.53
0.990	0.00	0.000	880.5	0.47805	0.00	0.07	8.53
1.049	0.00	0.000	882.0	0.47805	0.00	0.07	8.53
1.117	0.00	0.000	882.0	0.47805	0.00	0.07	8.53
1.188	0.00	0.000	880.5	0.47805	0.00	0.07	8.53
1.255	0.00	0.000	880.5	0.47805	0.00	0.07	8.53
1.479	0.00	0.000	880.5	0.47805	0.00	0.07	8.53
1.536	0.62	7.941	1407.5	0.47806	0.91	0.07	8.53
1.605	1.24	14.593	1407.5	0.47808	1.15	0.07	8.53
1.674	1.24	6.652	1407.5	0.47811	1.13	0.07	8.52
1.740	1.86	6.942	1717.0	0.47814	1.64	0.07	8.52
1.797	2.49	14.950	1717.0	0.47818	1.76	0.07	8.52
1.890	3.11	12.878	1717.0	0.47826	2.09	0.07	8.52
1.947	3.73	12.852	1881.0	0.47832	2.42	0.07	8.52
2.016	4.35	14.656	1881.0	0.47840	2.49	0.07	8.52
2.086	4.97	13.153	1881.0	0.47850	2.71	0.07	8.52
2.157	5.59	12.926	2001.5	0.47861	2.71	0.07	8.52
2.271	6.21	8.440	2001.5	0.47900	2.91	0.07	8.51
2.479	8.08	16.601	2110.0	0.47921	2.91	0.07	8.51
2.508	8.70	22.488	2299.5	0.47935	2.92	0.07	8.51

2.607	9.32	14.349	2299.5	0.47954	2.74	0.07	8.51
2.675	9.94	13.189	2299.5	0.47972	2.71	0.07	8.50
2.746	9.94	6.719	2560.5	0.47992	2.52	0.07	8.50
2.983	10.56	1.923	2560.5	0.48061	2.26	0.07	8.49
3.078	13.05	20.976	3004.5	0.48096	2.26	0.07	8.49
3.113	14.29	32.057	3451.5	0.48124	2.11	0.07	8.48
3.385	14.91	14.930	3451.5	0.48222	1.91	0.07	8.46
3.486	18.02	24.405	3964.0	0.48273	2.31	0.07	8.45
3.557	18.64	28.972	4522.5	0.48309	2.17	0.07	8.45
3.787	19.88	10.449	4522.5	0.48436	1.97	0.07	8.42
3.883	21.75	18.238	4992.5	0.48494	2.14	0.07	8.41
3.954	22.99	27.148	5348.5	0.48539	2.04	0.07	8.40
4.185	23.61	14.831	5348.5	0.48691	1.99	0.07	8.37
4.281	24.23	6.733	5423.0	0.48756	2.75	0.07	8.37
4.337	24.85	12.911	4507.0	0.48794	4.39	0.07	8.36
4.407	25.48	14.605	4507.0	0.48844	5.30	0.07	8.36
4.479	26.10	12.865	4507.0	0.48896	3.36	0.07	8.36
4.537	26.10	6.403	3791.0	0.48938	2.99	0.07	8.35
4.607	26.72	6.441	3791.0	0.48990	3.65	0.07	8.35
4.666	27.34	14.209	3791.0	0.49035	4.18	0.07	8.35
4.757	27.96	12.785	3793.0	0.49105	4.84	0.07	8.34
4.983	27.96	5.017	3793.0	0.49281	5.09	0.07	8.34
5.052	29.20	13.311	3931.0	0.49337	5.62	0.07	8.34
5.288	29.83	15.237	3931.0	0.49533	5.78	0.07	8.33
5.487	30.45	4.214	4186.5	0.49701	6.43	0.07	8.33
5.682	31.07	4.632	4151.5	0.49869	6.04	0.07	8.32
5.737	31.69	10.553	3497.0	0.49710	5.92	0.07	8.31
5.808	32.31	14.700	3497.0	0.49773	7.16	0.07	8.31
5.875	32.31	6.490	3497.0	0.49833	7.01	0.07	8.31
6.082	32.93	2.205	3154.0	0.50022	6.29	0.07	8.31
6.140	33.55	10.081	3132.0	0.50076	6.71	0.07	8.31
6.208	33.55	7.877	3132.0	0.50140	7.74	0.07	8.31
6.275	33.55	0.000	3132.0	0.50202	7.37	0.07	8.30
6.481	34.18	2.214	3095.5	0.50398	7.21	0.07	8.30
6.545	35.42	16.392	3122.5	0.50461	7.20	0.07	8.30
6.615	35.42	14.178	3122.5	0.50530	6.90	0.07	8.30
6.883	35.42	0.000	3122.5	0.50793	6.59	0.07	8.30
6.990	37.28	12.692	3219.5	0.50905	7.32	0.07	8.30
7.083	37.28	12.692	3219.5	0.51001	9.87	0.07	8.30
7.154	36.66	-6.459	3074.5	0.51073	12.00	0.07	8.30
7.383	36.66	-6.459	3074.5	0.51307	15.17	0.07	8.31
7.479	35.42	-9.541	3012.5	0.51401	29.55	0.07	8.31
7.587	34.80	-13.775	2941.5	0.51505	41.32	0.07	8.31
7.682	34.18	-8.998	2941.5	0.51595	48.31	0.07	8.32
7.740	33.55	-12.655	2804.0	0.51649	50.73	0.07	8.32
7.798	32.93	-15.772	2804.0	0.51702	53.77	0.07	8.32
7.996	32.31	-10.183	2804.0	0.51880	52.77	0.07	8.33
8.079	31.07	-13.247	2716.0	0.51952	51.01	0.07	8.34
8.187	31.07	-10.945	2633.0	0.52045	101.30	0.07	8.34
8.279	30.45	-4.985	2633.0	0.52122	109.83	0.07	8.34
8.336	29.83	-12.866	2501.5	0.52170	111.73	0.07	8.35
8.407	29.20	-14.344	2501.5	0.52227	111.73	0.07	8.35
8.497	29.20	-6.463	2501.5	0.52373	115.15	0.07	8.36
8.682	27.34	-14.309	2398.5	0.52446	107.80	0.07	8.36

8.778	26.72	-19.080	2286.0	0.52517	114.27	0.07	8.36
8.848	26.72	-4.771	2286.0	0.52569	119.89	0.07	8.37
8.906	26.10	-7.890	2286.0	0.52569	119.89	0.07	8.37
8.941	25.48	-14.371	2176.5	0.52661	114.68	0.07	8.37
9.423	24.85	-7.501	2176.5	0.52969	117.51	0.07	8.39
9.688	21.75	-9.625	1941.0	0.53129	102.82	0.07	8.40
9.881	20.51	-13.312	1741.5	0.53239	113.24	0.07	8.40
9.940	19.26	-20.364	1607.0	0.53270	113.82	0.07	8.40
9.997	18.64	-23.557	1607.0	0.53300	119.36	0.07	8.41
10.066	18.64	-7.900	1607.0	0.53336	119.36	0.07	8.41
10.283	18.02	-2.102	1517.0	0.53445	119.36	0.07	8.41
10.381	17.40	-4.420	1441.5	0.53540	118.38	0.07	8.42
10.584	15.53	-15.443	1341.0	0.53585	118.38	0.07	8.42
10.619	15.53	-13.124	1341.0	0.53615	119.57	0.07	8.42
10.724	14.91	-6.615	1341.0	0.53644	114.79	0.07	8.42
10.988	14.91	-6.615	1194.5	0.53753	114.79	0.07	8.43
11.082	13.67	-9.663	1052.5	0.53789	123.05	0.07	8.43
11.152	13.05	-16.244	1007.0	0.53814	139.10	0.07	8.43
11.210	12.43	-14.368	1007.0	0.53834	144.76	0.07	8.43
11.268	12.43	-7.787	1007.0	0.53854	138.46	0.07	8.44
11.326	12.43	0.000	908.5	0.53874	127.39	0.07	8.44
11.395	11.81	-6.556	908.5	0.53897	128.98	0.07	8.44
11.688	11.81	-6.556	908.5	0.53993	120.88	0.07	8.44
11.758	11.18	-6.453	862.0	0.54015	77.51	0.07	8.44
11.984	10.56	-8.474	862.0	0.54081	56.37	0.07	8.45
12.047	9.94	-9.204	850.0	0.54099	47.26	0.07	8.45
12.115	9.94	-7.183	850.0	0.54117	43.31	0.07	8.45
12.381	9.94	0.000	848.5	0.54191	41.39	0.07	8.45
12.439	9.32	-7.897	843.5	0.54206	37.09	0.07	8.45
12.497	9.32	-7.897	843.5	0.54221	34.83	0.07	8.45
12.568	9.32	0.000	854.0	0.54239	33.65	0.07	8.45
12.638	8.70	-6.464	854.0	0.54256	29.20	0.07	8.46
12.879	8.70	-6.464	854.0	0.54314	27.58	0.07	8.46
12.937	8.70	0.000	855.0	0.54328	23.29	0.07	8.46
13.007	8.08	-6.539	855.0	0.54344	20.52	0.07	8.46
13.077	8.08	-6.539	855.0	0.54360	19.68	0.07	8.46
13.147	8.08	0.000	863.5	0.54375	19.65	0.07	8.46
13.672	8.08	0.000	863.5	0.54493	19.44	0.07	8.46
13.729	7.46	-8.029	868.0	0.54505	17.93	0.07	8.46
13.796	7.46	-8.029	868.0	0.54519	17.85	0.07	8.46
13.886	6.84	-5.048	868.0	0.54536	16.04	0.07	8.46
13.954	6.84	-5.048	861.5	0.54549	16.04	0.07	8.46
14.184	6.21	-1.980	861.5	0.54589	14.69	0.07	8.47
14.401	5.59	-4.083	849.0	0.54622	12.97	0.07	8.47
14.783	3.73	-5.687	848.5	0.54662	8.77	0.07	8.47
14.853	1.24	-29.437	854.5	0.54664	3.06	0.07	8.47
15.080	1.24	-25.853	854.5	0.54672	3.23	0.07	8.47
15.139	0.00	-15.635	879.5	0.54672	3.23	0.07	8.47
15.168	0.00	-15.635	879.5	0.54672	0.00	0.07	8.46
15.267	0.00	0.000	879.5	0.54672	0.00	0.07	8.46
15.337	0.00	0.000	879.5	0.54672	0.00	0.07	8.46
15.580	0.00	0.000	879.5	0.54672	0.00	0.07	8.46
15.685	0.00	0.000	876.5	0.54672	0.00	0.07	8.46
15.755	0.00	0.000	879.5	0.54672	0.00	0.07	8.46

```
15.984     0.00        0.000     879.5    0.54672    0.00     0.07     8.46
16.079     0.00        0.000     876.5    0.54672    0.00     0.07     8.46
16.147     0.00        0.000     873.0    0.54672    0.00     0.07     8.46
16.379     0.00        0.000     873.0    0.54672    0.00     0.07     8.46
16.437     0.00        0.000     868.0    0.54672    0.00     0.07     8.46
16.508     0.00        0.000     868.0    0.54672    0.00     0.07     8.46
16.589     0.00        0.000     866.5    0.54672    0.00     0.07     8.46
16.647     0.00        0.000     866.5    0.54672    0.00     0.07     8.46
16.647     0.00        0.000     866.5    0.54672    0.00     0.07     8.46
Average sampling frequency for Jetta trials 1-3:[8.23,7.86,8.46] Hz
```

4. Helper Function Declaration

```
% this function takes in a string vector, and a cell
function parsedData = parsingData(newName,RawData)
    % the parsed data should be a cell too
    parsedData = cell(size(RawData));
    for i = 1:size(RawData,2)
       % the parsed data contains structs, same size as raw data cell
        parsedData{i} = struct();
        % for each element in the name field:
        for j = 1:numel(newName)
            % the chosen named struct object = corresponding raw data
            % column
            parsedData{i}.(newName(j)) = RawData{i}(:, j);
        end
    end
end

% this function calculate the average frequency of the data in each trial
function TimeData = calculateTimeData(timeVector)
    N = numel(timeVector);
    total_time = timeVector(end) - timeVector(1);
    TimeData.aveInterval = total_time/(N-1);
    TimeData.aveFrequency = 1/TimeData.aveInterval;
end

% this function prints 3 tables of data given a data cell containing 3
% trials
function table = printJettaDataTable(Datacell)
    for i = 1:size(Datacell,2)
        fprintf("\nJetta Trial %d\n",i);
        fprintf("%5s %4s %8s %3.5s %13s %7s %5s %5s\n", ...
    "Time(s)", "Speed(mph)", "Accel(ft/s^2)", ...
    "RPM", "TripDist(mi)", "InstMPG","TripFuel","TotalMPG");

        table = [Datacell{i}.Time_sec, Datacell{i}.VehicleSpeed_MPH, ...
                Datacell{i}.Acceleration_fts2, ...
                Datacell{i}.EngineRPM_rpm, ...
                Datacell{i}.TripDistance_mile, ...
                Datacell{i}.InstantFuelEcon_MPG,...
                Datacell{i}.TripFuel_gal,...
                Datacell{i}.TotalFuelEcon_MPG];
```

```
        fprintf("%5.3f %8.2f %12.3f %9.1f %10.5f %7.2f " + ...
            "%8.2f %8.2f\n", table.');
    end
end

% prints the sampling frequency of the 3 trials
fprintf("Average sampling frequency for Jetta trials 1-3:" + ...
    "[%.2f,%.2f,%.2f] Hz\n",[JettaTimeData{1}.aveFrequency,...
    JettaTimeData{2}.aveFrequency,...
    JettaTimeData{3}.aveFrequency]);
```

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